SM&I - Test 1

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Question 1

R Script and Output

```
# Question 1: Two-sample inference for porosity means

n1 <- 40; ybar1 <- 0.22; s1_sq <- 0.0010
n2 <- 35; ybar2 <- 0.17; s2_sq <- 0.0020

diff <- ybar1 - ybar2
SE <- sqrt(s1_sq/n1 + s2_sq/n2)

# Welch-Satterthwaite degrees of freedom
df_num <- (s1_sq/n1 + s2_sq/n2)^2
df_den <- (s1_sq/n1)^2/(n1-1) + (s2_sq/n2)^2/(n2-1)
df <- df_num / df_den

# 95% CI
t_crit <- qt(0.975, df)
CI_lower <- diff - t_crit * SE
CI_upper <- diff + t_crit * SE

# One-sided test H0: mu1-mu2 <= 0.02 vs H1: > 0.02
delta0 <- 0.02
t_stat <- (diff - delta0) / SE
p_value <- 1 - pt(t_stat, df)

cat("Point estimate =", diff, "\n")
cat("SE =", SE, "\n")
cat("df =", df, "\n")
cat("95% CI: (", CI_lower, ",", CI_upper, ")\n")
cat("t-statistic =", t_stat, "\n")
cat("One-tailed p-value =", p_value, "\n")
if (p_value < 0.05) cat("Reject H0: mu1 - mu2 > 0.02\n")
```

Console Output:

```
Point estimate = 0.05
SE = 0.00906327
df = 60.21064
95% CI: ( 0.03187207 , 0.06812793 )
t-statistic = 3.310064
One-tailed p-value = 0.0007894957
Reject H0: mu1 - mu2 > 0.02
```

Answers and Interpretation

(a) 95% Confidence Interval for $\mu_1 - \mu_2$

Using the Welch two-sample t-interval (unequal variances), the estimated difference is $\hat{\mu}_1 - \hat{\mu}_2 = 0.05$ with standard error 0.00906 and 60.2 degrees of freedom. The 95% confidence interval is

$$(0.0319,\ 0.0681).$$

We are 95% confident that the true difference in mean porosity (lab1 – lab2) lies between 0.0319 and 0.0681.

(b) Hypothesis Test

We test

$$H_0 : \mu_1 - \mu_2 \leq 0.02 \quad \text{vs.} \quad H_1 : \mu_1 - \mu_2 > 0.02$$

at the 5% significance level. The test statistic is $t = 3.310$ with 60.2 df, giving a one-tailed p-value of 0.0008. Since $p < 0.05$, we reject H_0. There is sufficient evidence to conclude that the difference between the population means is greater than 0.02.

(c) Interpretation

The 95% confidence interval lies entirely above 0.02 (lower bound $0.0319 > 0.02$), which aligns with the rejection of H_0 in part (b). Thus, laboratory1 produces solid copper with a statistically significantly higher mean porosity than laboratory2, and the difference exceeds 0.02 at the 5% significance level.

Question 2

R Script and Output

```
# Question 2: Linear regression of Horsepower on Cylinder Volume

cyl_vol <- c(1.8, 1.5, 2.0, 2.5, 1.8, 2.5, 1.6, 1.5)
horsepower <- c(51, 51, 115, 150, 126, 150, 118, 106)

model <- lm(horsepower ~ cyl_vol)
summary(model)

# Scatter plot with regression line
plot(cyl_vol, horsepower,
     xlab = "Cylinder Volume (L)",
     ylab = "Horsepower",
     main = "Horsepower vs. Engine Size")
abline(model, col = "red", lwd = 2)

# Estimate mean horsepower for 1.9 L
new_data <- data.frame(cyl_vol = 1.9)
predict(model, new_data, interval = "confidence", level = 0.95)
```

Console Output:

```
Call:
lm(formula = horsepower ~ cyl_vol)

Residuals:
    Min      1Q  Median      3Q     Max
-50.858  -7.746   2.522  23.806  29.177

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept)    -15.45       54.84   -0.282    0.7876
cyl_vol         65.17       28.30    2.303    0.0609 .
---
Signif. codes:  0     ***     0.001     **     0.01     *     0.05     .
   0.1          1

Residual standard error: 30.48 on 6 degrees of freedom
Multiple R-squared:  0.4692,    Adjusted R-squared:  0.3807
F-statistic: 5.303 on 1 and 6 DF,  p-value: 0.06087

      fit      lwr      upr
1 108.375 82.00477 134.7452
```

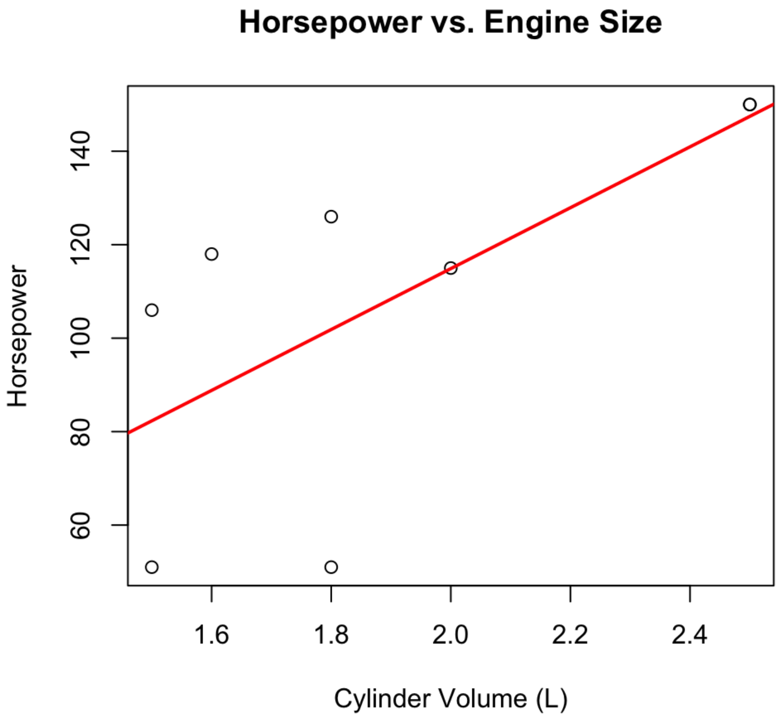

Figure 1: Scatter plot of Horsepower vs. Cylinder Volume with the least-squares line.

Answers

(a) Least-squares line

Based on the R output, the estimated intercept $b_0 = -15.45$ and slope $b_1 = 65.17$. The regression equation is

$$\widehat{\text{Horsepower}} = -15.45 + 65.17 \times \text{Cylinder Volume (L)}.$$

(b) Graph

The scatter plot (Figure1) shows a moderate positive linear relationship. The least-squares line (red) captures the general upward trend, but the fit is not very strong: the residual standard error is 30.48, and $R^2 = 0.4692$ indicates that only about 46.9% of the variability in horsepower is explained by cylinder volume. The p-value for the slope is 0.0609, which is marginally above 0.05, suggesting that the relationship is not statistically significant at the conventional 5% level.

(c) Estimated mean horsepower for 1.9L

For a cylinder volume of 1.9L, the predicted mean horsepower is 108.38. The 95% confidence interval for the true mean horsepower of all such automobiles is (82.00, 134.75).